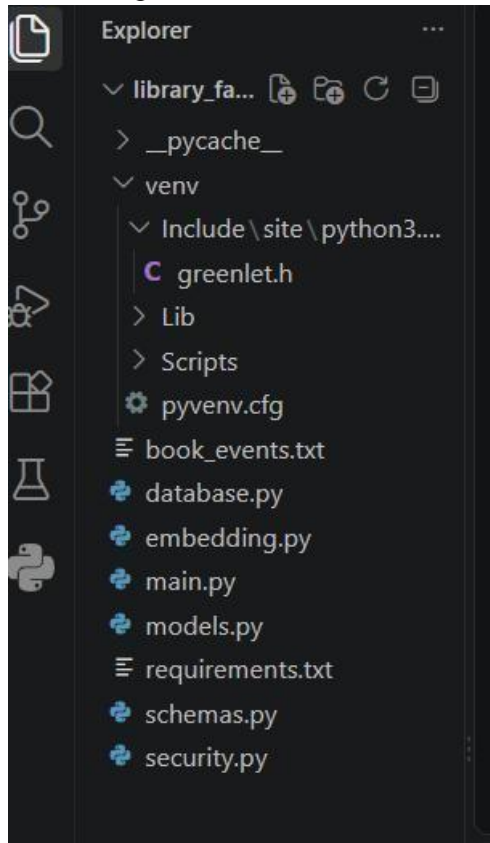


Practical-5

2520090198-Siri Chandana

1. Project Structure



2. Installed Packages

```

File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\starlette\routing.py", line 648, in lifespan
    async with self.lifespan_context(app) as maybe_state:
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 265, in __aenter__
    await self.router.startup()
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 6375, in _startup
    await handler()
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\main.py", line 38, in startup
    async with engine.begin() as connection:
File "C:\Users\satwika\AppData\Local\Programs\Python\Python311\Lib\contextlib.py", line 210, in __aenter__
    return await anext(self.gen)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\engine.py", line 1068, in
    begin with conn:
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\base.py", line 121, in
    in __aenter__:
    return await self.start(is_ctxmanager=True)

File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 144, in __init__
    self._dbapi_connection = engine.raw_connection()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 3319, in raw_connection
    return self.pool.connect()
    ^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 448, in connect
    return _ConnectionFairy._checkout(self)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 1272, in _checkout
    fairy = _ConnectionRecord.checkout(pool)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 712, in checkout
    rec = pool._do_get()
    ^^^^^^^^^^^^^^^^^^^^^^^^^

```

•

```

File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\starlette\routing.py", line 648, in lifespan
    async with self.lifespan_context(app) as maybe_state:
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 265, in __aenter__
    await self.router.startup()
    ^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\main.py", line 38, in startup
    async with engine.begin() as connection:
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 1272, in _enter_
    return await _create_connection()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 712, in _checkout
    rec = pool._do_get()
    ^^^^^^^^^^^^^^^^^^^^^^^^^

```

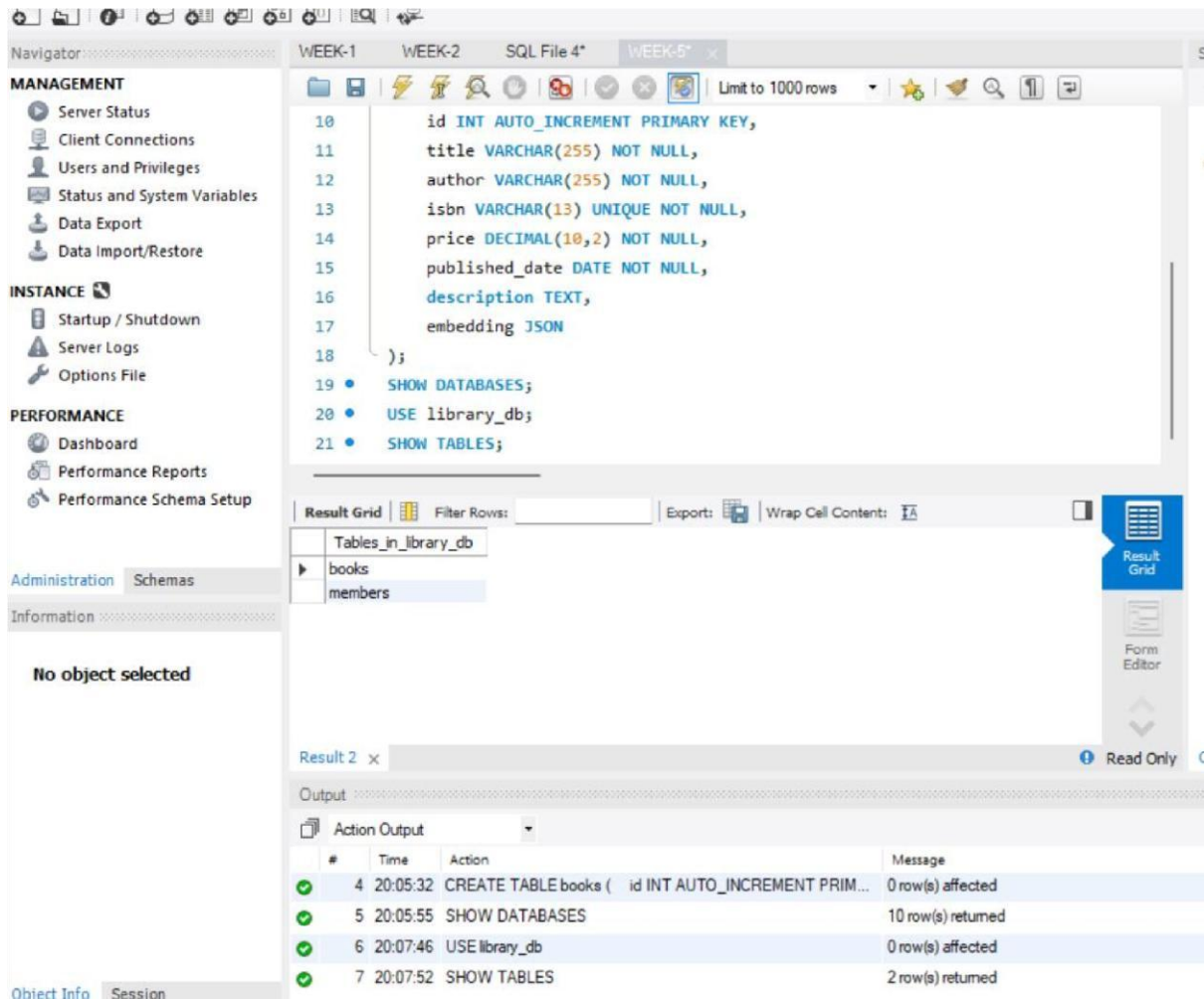
The screenshot displays the MySQL Workbench interface. The top toolbar includes icons for SQL, Schema, Data, and other database operations. The left sidebar shows the 'MANAGEMENT' and 'INSTANCE' sections. The main editor window shows a SQL script with the following queries:

```

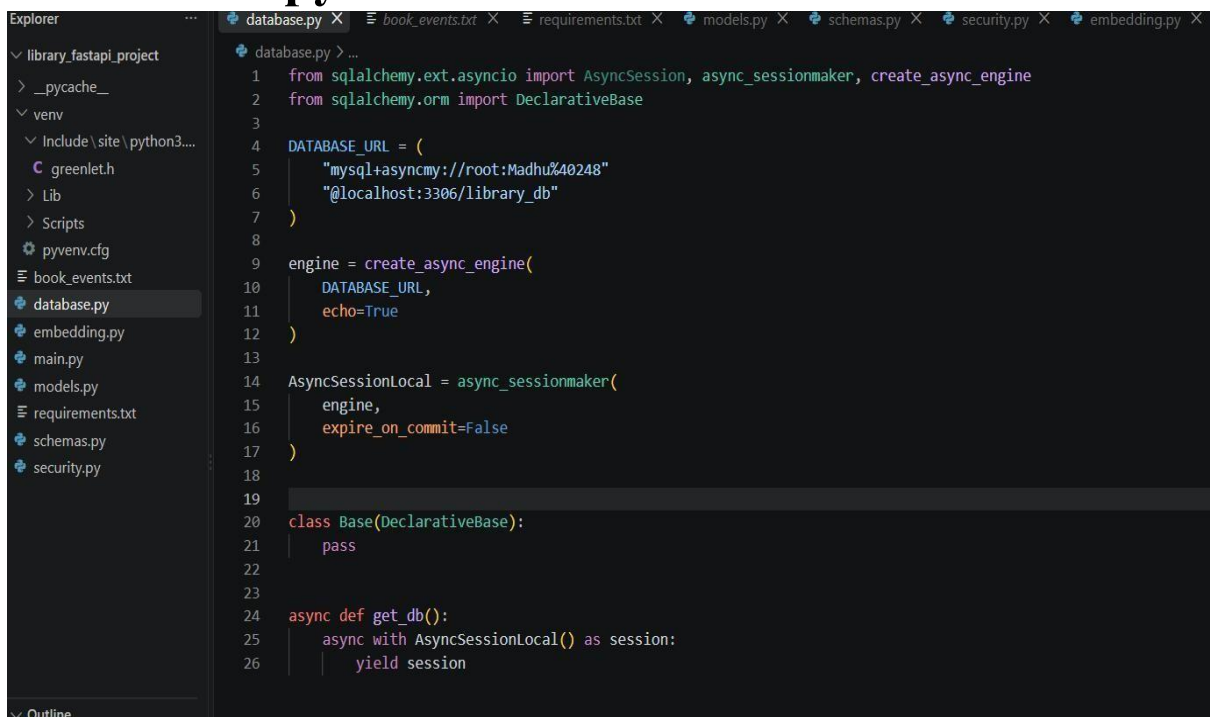
1 • CREATE DATABASE library_db;
2 • USE library_db;
3 • CREATE TABLE members (
4     id INT AUTO_INCREMENT PRIMARY KEY,
5     username VARCHAR(100) UNIQUE NOT NULL,
6     password_hash VARCHAR(255) NOT NULL,
7     role VARCHAR(30) DEFAULT 'member'
8 );

```

The 'Result Grid' is visible at the bottom, showing the output of the queries. The first query, 'SHOW DATABASES;', returns a single row with the database name 'bookflow_db'. The second query, 'USE library_db;', returns a message '0 row(s) affected'. The third query, 'CREATE TABLE members (...)', returns a message '0 row(s) affected'. The fourth query, 'CREATE TABLE books (...)', returns a message '0 row(s) affected'. The fifth query, 'SHOW DATABASES;', returns 10 rows, including 'bookflow_db', 'bookflow_dh', 'information_schema', 'kludb', 'library_db', 'madhulka_2520090056', 'mysql', and 'performance_schema'.



4. Database.py



5. Embedding.py

```
library_fastapi_project
├── __pycache__
├── venv
│   ├── Include\site\python3...
│   ├── greenlet.h
│   ├── Lib
│   ├── Scripts
│   ├── pyvenv.cfg
│   ├── book_events.txt
│   ├── database.py
│   ├── embedding.py
│   ├── main.py
│   ├── models.py
│   ├── requirements.txt
│   ├── schemas.py
│   └── security.py
└── ...

embedding.py > text_to_vector
1  import hashlib
2  import math
3
4
5  VECTOR_DIM = 384
6
7
8  def text_to_vector(text: str):
9      vector = []
10
11      for i in range(VECTOR_DIM):
12          h = hashlib.sha256(
13              f"{text}-{i}".encode()
14          ).digest()
15
16          value = int.from_bytes(h[:4], "big") / (2**32 - 1)
17          vector.append(value)
18
19      norm = math.sqrt(
20          sum(x * x for x in vector)
21      )
22
23      if norm == 0:
24          return vector
25
26      return [x / norm for x in vector]
```

6. Main.py

```
library_fa...
├── __pycache__
├── venv
│   ├── Include\site\python3...
│   ├── greenlet.h
│   ├── Lib
│   ├── Scripts
│   ├── pyvenv.cfg
│   ├── book_events.txt
│   ├── database.py
│   ├── embedding.py
│   ├── main.py
│   ├── models.py
│   ├── requirements.txt
│   ├── schemas.py
│   └── security.py
└── ...

main.py > login
1  from fastapi import (
2      BackgroundTasks,
3      Depends,
4      FastAPI,
5      HTTPException,
6      Query,
7      status
8  )
9
10 from fastapi.security import OAuth2PasswordRequestForm
11
12 from sqlalchemy import select
13 from sqlalchemy.ext.asyncio import AsyncSession
14
15 from database import Base, engine, get_db
16 from embedding import text_to_vector
17 from models import Book, Member
18 from schemas import BookSchema, MemberRegister, TokenResponse
19
20 from security import (
21     create_access_token,
22     get_current_member,
23     hash_password,
24     verify_password
25 )
26
27 import math
28
29
30 app = FastAPI(
31     title="Library Backend & Security API",
32     version="1.0"
33 )
34
35
36 @app.on_event("startup")
```

7. Models.py

```
library_fastapi_project
├── __pycache__
├── venv
├── Include\site\python3....
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
└── security.py
```

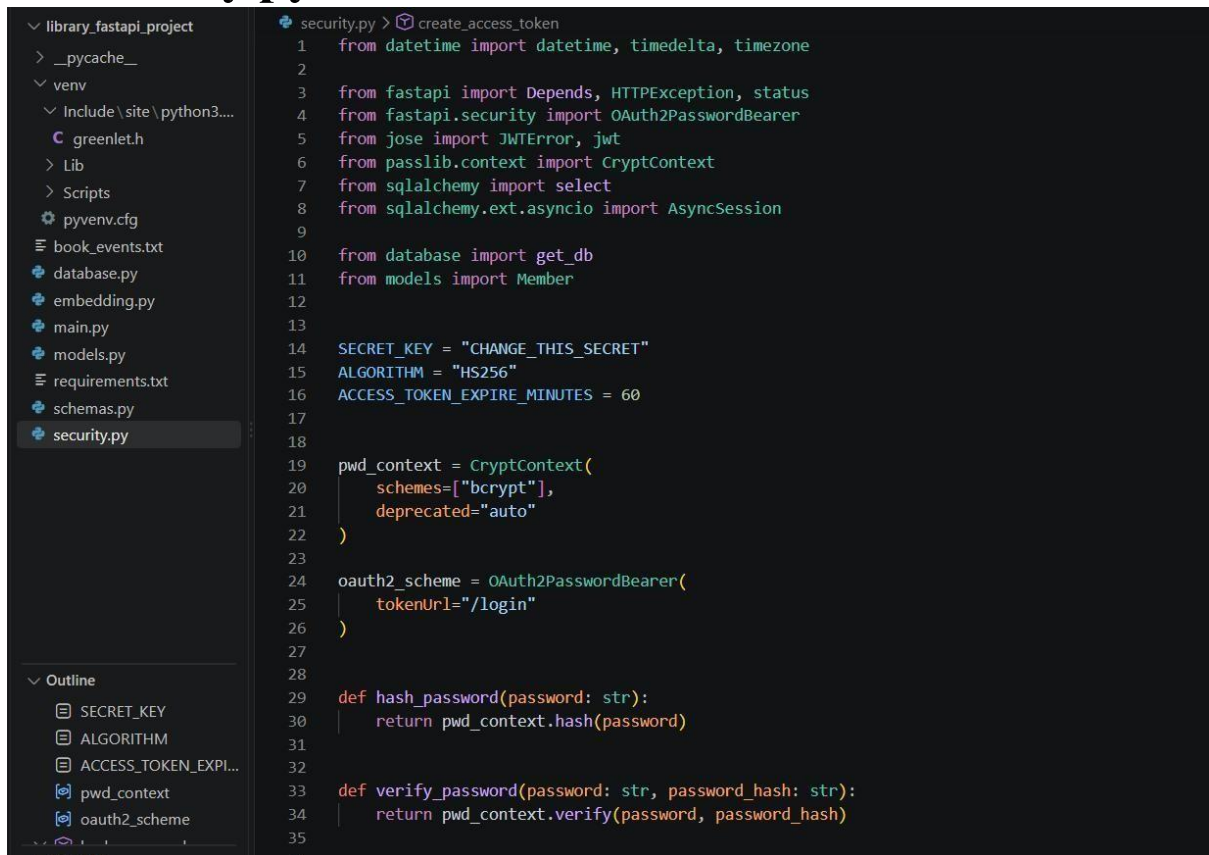
```
models.py > Book
1  from sqlalchemy import Column, Integer, String, Text, DECIMAL, Date, JSON
2  from database import Base
3
4
5  class Member(Base):
6      __tablename__ = "members"
7
8      id = Column(Integer, primary_key=True, index=True)
9      username = Column(String(100), unique=True, nullable=False)
10     password_hash = Column(String(255), nullable=False)
11     role = Column(String(30), default="member")
12
13
14     class Book(Base):
15         __tablename__ = "books"
16
17         id = Column(Integer, primary_key=True, index=True)
18         title = Column(String(255), nullable=False)
19         author = Column(String(255), nullable=False)
20         isbn = Column(String(13), unique=True, nullable=False)
21         price = Column(DECIMAL(10, 2), nullable=False)
22         published_date = Column(Date, nullable=False)
23         description = Column(Text)
24         embedding = Column(JSON)
```

8. Schemas.py

```
Explorer
├── library_fastapi_project
├── __pycache__
├── venv
├── Include\site\python3....
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
└── security.py
```

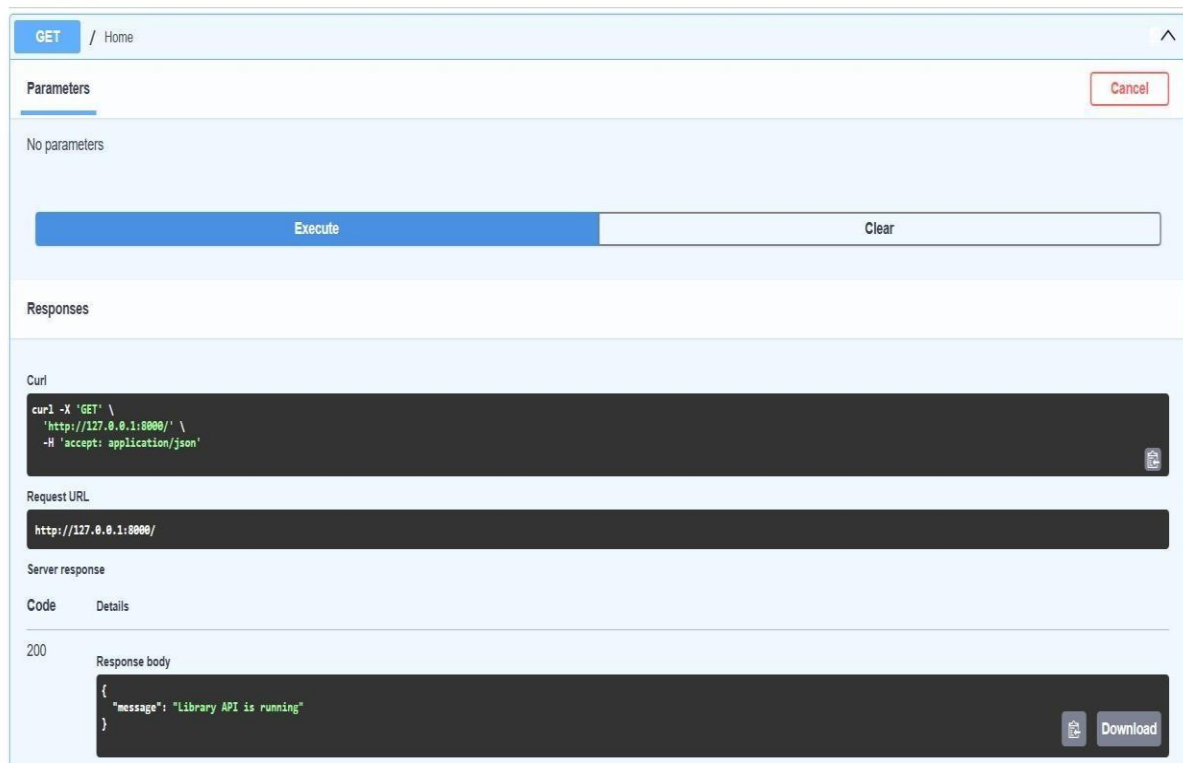
```
schemas.py > MemberRegister
1  from pydantic import BaseModel, Field, field_validator
2  from datetime import date
3
4
5  class BookSchema(BaseModel):
6      title: str
7      author: str
8      isbn: str = Field(min_length=13, max_length=13)
9      price: float = Field(gt=0)
10     published_date: date
11     description: str | None = None
12
13     @field_validator("published_date")
14     @classmethod
15     def validate_published_date(cls, value):
16         if value > date.today():
17             raise ValueError("published_date cannot be in the future")
18         return value
19
20
21     class MemberRegister(BaseModel):
22         username: str
23         password: str = Field(min_length=6)
24
25
26     class TokenResponse(BaseModel):
27         access_token: str
28         token_type: str
```

9. Security.py



```
security.py > create_access_token
1  from datetime import datetime, timedelta, timezone
2
3  from fastapi import Depends, HTTPException, status
4  from fastapi.security import OAuth2PasswordBearer
5  from jose import JWTError, jwt
6  from passlib.context import CryptContext
7  from sqlalchemy import select
8  from sqlalchemy.ext.asyncio import AsyncSession
9
10 from database import get_db
11 from models import Member
12
13
14 SECRET_KEY = "CHANGE_THIS_SECRET"
15 ALGORITHM = "HS256"
16 ACCESS_TOKEN_EXPIRE_MINUTES = 60
17
18
19 pwd_context = CryptContext(
20     schemes=["bcrypt"],
21     deprecated="auto"
22 )
23
24 oauth2_scheme = OAuth2PasswordBearer(
25     tokenUrl="/login"
26 )
27
28
29 def hash_password(password: str):
30     return pwd_context.hash(password)
31
32
33 def verify_password(password: str, password_hash: str):
34     return pwd_context.verify(password, password_hash)
35
36
```

10. Test Registration



GET / Home

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body <pre>{ "message": "Library API is running" }</pre>

Download

POST/registerRegister

Parameters

CancelReset

No parameters

Request bodyrequired

application/json

Edit ValueSchema

```
{
  "username": "Madhu",
  "password": "library123"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Madhu",
    "password": "library123"
  }'
```

Request URL

http://127.0.0.1:8080/register

Server response

CodeDetails

200

Response body

```
{
  "message": "Member registered successfully",
  "username": "Madhu",
  "role": "member"
}
```

Download

Response headers

```
content-length: 79
content-type: application/json
date: Mon, 14 Sep 2026 16:05:16 GMT
server: uvicorn
```

Responses

11. Test Login

POST/login Login

Parameters

CancelReset

No parameters

Request body required

application/x-www-form-urlencoded

grant_type

string | (string | null)
pattern: "password\$"

password

☐ Send empty value

username * required

string

Madhu

password * required

string(\$password)

scope

string

scope

☒ Send empty value

client_id

string | (string | null)

Madhu

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/login' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=Madhu&password=library123&scope=&client_id=Madhu&client_secret=library123'
```

Request URL

http://127.0.0.1:8000/login

Server response

Code

Details

200

Response body

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IWRXVj09LWVudGUiOiJ0eXN0aWZlIiwiaWF0IjE5ODkxODU5Nzh9.BSylTtpIyOmAoCodhwV3PUgLvVb9I2uZ8d8yfyJVx4",
  "token_type": "bearer"
}
```

CopyDownload

Response headers

```
content-length: 186
content-type: application/json
date: Mon, 14 Sep 2026 16:12:57 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json

Controls Accept header

Example Value | Schema

12. Test POST/Books

POST

/books Create Book

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "title": "Python for Beginners",
  "author": "John Smith",
  "isbn": "123",
  "price": -100.00
  "published_date": "2024-06-10",
  "description": "An introduction to Python programming."
}
```

Execute

Clear

Responses

```
curl -X POST \
  http://127.0.0.1:8080/books \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpVC39.ej7dWl0iJWVwRodStInJvYmU0Ij7ZWI2XTI1CjE1eHA0IjE3ODk0MDVmMjY9.Lq7z8UsTHNKKyQbXBxEGTX64tGklS-Hhu4Jg@HnB110' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Python for Beginners",
    "author": "John Smith",
    "isbn": "123",
    "price": 450.00,
    "published_date": "2024-06-10",
    "description": "An introduction to Python programming."
  }'
```

Request URL

```
http://127.0.0.1:8000/books
```

Server response

Code	Details
------	---------

422 Error: Unprocessable Entity

Response body

```
{
  "detail": {
    {
      "type": "string_too_short",
      "loc": {
        "body",
        "isbn"
      },
      "msg": "String should have at least 13 characters",
      "input": "123",
      "ctx": {
        "min_length": 13
      }
    }
  }
}
```

Response headers

```
content-length: 150
content-type: application/json
date: Mon, 14 Sep 2026 16:20:41 GMT
server: uvicorn
```

GET
/books/search
Search Books

Parameters
Cancel

Name	Description
q * required string (query)	q = python minLength: 1
limit integer (query)	5 maximum: 20 minimum: 1

Execute
Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books/search?q=q%20python&limit=5' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books/search?q=q%20python&limit=5
```

Server response

13. book_event.txt

library_fastapi_project
__pycache__
venv
Include\site\python3....
greenlet.h
Lib
Scripts
pyenv.cfg
book_events.txt
database.py
embedding.py
main.py
models.py
requirements.txt
schemas.py
security.py

book_events.txt

```

1 New Book Added: id=1, title=Python for Beginners
2

```

